Department of Computer Science and Engineering
UIET, CSJM University, Kanpur

C Programming Laboratory Assignment–3

My First C Programs: Variables and `printf`

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What You Will Practise

Writing a basic C program in gVim, declaring and initializing variables, displaying text and variable values with **`printf`**, compiling with **`gcc`**, running the executable, and correcting simple errors.

Quick Revision

`gvim filename.c`	Create or edit a C source file.
`gcc filename.c -o name`	Compile it and create an executable.
`./name`	Run the executable in the current directory.
`int`, `float`, `char`	Store an integer, fractional value or character.
`%d`, `%f`, `%c`	Display an `int`, `float` or `char`.
`\n`	Move the following output to a new line.

Important Rules

Work inside `~/C_Programming/Lab_03`. Save every source file with the `.c` extension. Compile after every change. Read the first compiler error carefully. Do not copy another student's program.

Preparation

Create **`Lab_03`** inside your C-programming lab directory. Enter it and confirm the location using **`pwd`**. All three programs must be saved here.

Problem 1: Your UIET Digital Introduction

Situation

The computer laboratory needs a simple program that displays a student's basic UIET information. The values are already stored in the program; no keyboard input is required.

Create **`student.c`**. Declare and initialize variables for your roll number, semester number, section, and first-semester percentage. Display a neat output similar to this:

```
----- UIET STUDENT PROFILE -----
Roll number : 2401234
Semester    : 1
Section     : A
Percentage  : 78.50
Learning C programming at CSJM University!
```

1. Use suitable variables of type **`int`**, **`float`** and **`char`**.
2. Use one or more **`printf`** statements to produce the output.
3. Display the percentage with exactly two digits after the decimal point.
4. Compile it as **`student`** and run it.
5. Change the values to your own details, save, compile and run again.

Be ready to explain: the purpose of **`#include <stdio.h>`**, **`main`**, each variable, each format specification and **`return 0;`**.

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Problem 2: Register a UIET Laboratory Computer

Situation

UIET is preparing a display program for the computers in its programming laboratory. Your program will store a machine's details and print its record.

Create `computer.c`. Store the following information in variables:

- laboratory number: `3`; computer number: `27`;
- RAM in GB: `16`; free disk space in GB: `125.75`;
- computer status: character `'W'` for working.

Print a labelled record headed `UIET C PROGRAMMING LAB`. Then:

1. compile and run the program;
2. change the computer number from `27` to `28`;
3. change the free disk space to `120.50`;
4. change the status from `'W'` to `'M'` for maintenance;
5. save, rebuild and run; check that all three changes appear.

Think Before You Type

Which type and **`printf`** format will you use for every value? Remember that a character uses single quotation marks, for example `'W'`.

Problem 3: Build a Campus Utility Information Screen

Situation

Imagine that your program will later become part of a UIET Campus Utility system. For now, it only has to store fixed values and display them clearly.

Create `campus.c`. Declare and initialize variables for:

- available library seats: `42`;
- canteen token number: `108`;
- tea price: `12.50`;
- attendance percentage: `86.75`;
- block letter: `'C'`.

Use **`printf`** to make a readable information screen. Include a heading, labels, units such as `rupees` and `percent`, and a closing message. Use `\n` to arrange the output on separate lines.

After the correct run, make these experiments one at a time. Predict what will happen, compile, observe the message, and then repair the program:

1. remove one semicolon;
2. misspell **`printf`** as **`print`**;
3. remove the closing double quote from one message;
4. use **`%d`** for the tea-price variable;
5. change a variable's value but run the old executable without compiling.

Record the first compiler warning or error from any two experiments in your notebook and write, in simple words, how you corrected it.

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Final Challenge and Demonstration

Without copying an earlier program, create **about_me.c**. It must contain at least four variables, use at least three different data types, and display a five-line introduction about you as a UIET student. Give variables meaningful names and format the output neatly.

Show Before Leaving the Lab

Show the instructor the four source files and their outputs. Demonstrate how you edit one value, save the file, compile it again, and obtain the changed output. Be prepared to explain one declaration and one **printf** statement.

Remember: source file → compiler → executable → output.

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